

How Many More Labels Are Needed? Minimax Evaluation under Selective Labels

Submitted to ICLR 2027 · Decision pending (not accepted) | Summary: 27 September 2026

How many additional labels are needed to detect unacceptable risk when the labels already observed may be selectively missing?

CENTRAL IDEA

Treat evaluation as a minimax audit of a frozen cohort. An auditor privately queries committed outcomes and must control false alarms even when the original missingness depends on the outcome.

MAIN FINDING

The paper characterizes exact fixed-query asymptotic power with truthful labels, then separates noisy-label guarantees that look similar but induce different testing problems. Robust frontiers translate uncertainty about label quality into query requirements and blind regions.

EVIDENCE AND SCOPE

In one scoped finite design, distinguishing 10% from 20% risk at 5% false-alarm probability and 95% power requires 57 queries among 1,000 predictions with 500 pending labels. Detecting 12% risk is limited by cohort size: even complete labels require at least 2,645 predictions in that design.

LIMITS

The numerical designs are not a universal finite-budget formula. Results require the stated cohort, query and label-reliability assumptions. Failure to reject unacceptable risk does not certify safety.

TAKEAWAY

Audit budgets depend on three resources: the number of queries, knowledge of label quality, and the size of the evaluation cohort.

한국어 핵심 요약

관측된 라벨이 좋은 사례에 편중되어 있을 때, 추가 라벨 몇 개로 위험 증가를 발견할 수 있는지 연구합니다. 필요한 예산은 질의 수뿐 아니라 라벨의 신뢰도와 평가 집단 크기에 따라 달라집니다. 정해진 가정 아래의 최적 검정력과 한계를 제시하며, 유의하지 않은 결과를 안전성 인증으로 해석하지 않습니다.

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